

ECON-ARK: OPEN-SOURCE TOOLS FOR HETEROGENEOUS AGENT MODELING

Advancing Economic Research & Industry Applications

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<https://econ-ark.org>

Introduction & Motivation

- Econ-ARK is an open-source toolkit for **heterogeneous agent modeling**.
- Principal Investigator: Prof. Chris Carroll (JHU) — known for groundbreaking work on **consumption** and the **Endogenous Grid Method (EGM)**.
- Presenter: Alan Lujan, PhD (JHU) — Core Developer & Maintainer of Econ-ARK.
- Fiscal Sponsor: NumFOCUS — Corporate Sponsor: T. Rowe Price (TRP)
- **Goal:** Reduce coding overhead so economists, students, and policymakers, can focus on **economic insights** — spanning **macro**, **micro**, and **policy** questions.

Econ-ARK Overview

- **Vision:** Provide a unified platform for **modeling, simulating, and analyzing** heterogeneous individual behavior and its **aggregate implications**.
- **Open-Source:** Encourages transparency, reproducibility, and fosters a growing community.
- **Python-Based:** Leverages the scientific computing ecosystem (NumPy, SciPy, Pandas, etc.).
- **Modular & Extensible:** Easily adapt or extend existing models; share solutions across academia and industry.

HARK: The Core Python Library

- **HARK (Heterogeneous Agents Resources and toolKit):**
 - Offers **pre-built** agent models (e.g., consumption-saving, portfolio choice, **lifecycle**).
 - Implements an **object-oriented** approach for defining and simulating new agent types.
 - Based on and expands **Chris Carroll's** pioneering methods (e.g., **EGM** for solving consumption problems).
- **REMARK (Replications and Explorations Made using ARK):**
 - Bundled Jupyter notebooks, data, and instructions enable **replicable** research and teaching.
 - Ideal for sharing open research workflows.

Key Academic Use Cases

- **Structural Modeling:** Rapid experimentation with established or novel frameworks (buffer-stock saving, consumption-savings models, etc.).
- **Heterogeneous Agent New Keynesian (HANK):** Analyze policy experiments under rich micro-level heterogeneity.
- **Lifecycle Modeling:** Incorporate realistic income processes, demographics, retirement choices, and policy uncertainty.
- **Facilitating Cutting-Edge Research:** Focus on the economics rather than building modeling infrastructure from scratch.
- **Open Science:** Encourage collaboration and reproducibility.

Key Industry Use Cases

- **Financial Services:** Personalized wealth management, retirement advice, and product design.
- **Risk Analytics:** Scenario analysis for insurance, mortgage, and consumer credit markets.
- **Policy Evaluation:** Model macro-level outcomes (e.g., consumption, default risk, wealth distribution) from heterogeneous micro-level behavior.
- **Rapid Prototyping:** Adapt academic-grade models to real-world data and scale them for commercial applications.
- **Forecasting:** Use macro models to predict future economic conditions and policy decisions.

Example: Consumption-Saving over the Life Cycle

A **consumption-saving** model with persistent income shocks, where all variables are “normalized” (i.e., **divided by permanent income**).

$$v_t(m_t) = \max_{c_t} \left\{ u(c_t) + \beta \mathbb{E}_t \left[(\Gamma_{t+1} \psi_{t+1})^{1-\rho} v_{t+1}(m_{t+1}) \right] \right\}$$

$$m_{t+1} = \frac{R_{t+1}}{\Gamma_{t+1} \psi_{t+1}} (m_t - c_t) + \theta_{t+1}$$

- m_t : normalized wealth
- c_t : consumption choice
- $\Gamma_{t+1}, \psi_{t+1}, \theta_{t+1}$: growth factors & shocks
- R_{t+1} : risky return on savings

Quick Demo

```
1 from HARK.ConsumptionSaving.ConsIndShockModel import IndShockConsumerType
2
3 ## This code defines, solves, and simulates a basic consumption-saving model
4
5 ## Define parameters
6 params = {
7     'CRRA': 2.0,
8     'Rfree': 1.03,
9     'DiscFac': 0.96,
10    'PermShkStd': [0.1],
11    'TranShkStd': [0.1],
12    'cycles': 0 ## Infinite horizon for demo
13 }
14
15 ## Create and solve the consumer instance
16 consumer = IndShockConsumerType(**params)
17 consumer.solve()
18 consumer.simulate(50) ## Simulate 50 periods
```

- **Takeaway:** HARK handles much of the infrastructure for defining, solving, and simulating agent-based models.

Open Science & Community

- **REMARKs (Replications and Explorations Made using ARK):**
 - Integrated notebooks, datasets, instructions for fully reproducible projects.
- **Get Involved:**
 - Contribute ideas, report issues, or submit pull requests on **GitHub**.
 - Build and share new models or improvements with a global community.
- **Events & Collaboration:**
 - Community calls, workshops, and conferences to discuss use cases and best practices.

Key Takeaways

1. **Econ-ARK** accelerates the development of **heterogeneous agent models** for research and industry.
2. **HARK** integrates **Chris Carroll's** EGM approach and supports structural modeling, including **HANK** and **lifecycle** frameworks.
3. **Open-source** fosters transparency, reproducibility, and an engaged community.

We're always looking for collaborators, partners, and additional corporate sponsors for advanced modeling and applications!

- **Website:** econ-ark.org
- **Repository:** github.com/econ-ark/HARK
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